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I confirm that I understand my coursework needs to be submitted online via Google Classroom under the relevant module page before the deadline in order for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded.

Table of Contents

Logbook 8 1
1.1 Aim 1
1.2 Objectives 1
1.3 Required tools and concepts 2
1.4 Steps required for lab 3
1.5 Conclusion 20

Table of Figures

Figure 1: Creating directory structure 3
Figure 2: Creating directory structure with -p option 4
Figure 3: changing directory to 1level3 4
Figure 4: Changing directory 4
Figure 5: Creating a text file 5
Figure 6: Copying file to different directories 6
Figure 7: Moving file to another directory 6
Figure 8: Printing the text using echo 7
Figure 9: using ls in w7 directory 7
Figure 10: Using ls in w7 with option "a" 7
Figure 11: Using ls in w7 with option "d" 7
Figure 12: Using ls in w7 with option "g" 8
Figure 13: Using ls in w7 with option "l" 8
Figure 14: Using ls in w7 with option "R" 9
Figure 15: Using ls in w7-1 directory 10
Figure 16: Using ls in w7-1 with option "a" 10
Figure 17: Using ls in w7-1 with option "d" 10
Figure 18: Using ls in w7-1 with option "g" 10
Figure 19: Using ls in w7-1 with option "l" 10
Figure 20: Using ls in w7-1 with option "R" 11
Figure 21: Using ls in 1level3 directory 11
Figure 22: Using ls in 1level3 with option "a" 11
Figure 23: Using ls in 1level3 with option "d" 12
Figure 24: Using ls in 1level3 with option “g” 12
Figure 25: Using ls in 1level3 with option “l” 12
Figure 26: Using ls in 1level3 with option "R" 12
Figure 27: Changing directory to w7 13
Figure 28: Removing w7-2 13
Figure 29: Changing to w7-1 directory 14
Figure 30: Displaying the access permissions 14

Figure 31: Removing the access permission 14
Figure 32: Reading the file 15
Figure 33: Writing in the file 15
Figure 34: Adding and displaying the permissions 15
Figure 35: Reading the file after giving permission 16
Figure 36: Writing in the file after giving permission 16
Figure 37: Displaying the file where we made change 16
Figure 38: Displaying the file permissions 16
Figure 39: Removing and displaying access permission 17
Figure 40: Reading the file 17
Figure 41: Creating a file using touch 17
Figure 42: Listing what is inside 1level3 18
Figure 43: Adding read, write, and execute access 18
Figure 44: Displaying the access permission of 1level3 18
Figure 45: Reading file from 1level3 after giving access to user 19
Figure 46: Creating file inside 1level3 19
Figure 47: Searching file inside 1level3 19

Logbook 8

1.1 Aim

The aim of this lab is to learn how to create a directory and its subdirectories, execute file operations between directories, practice the variations of ls commands, test and change file permissions.

1.2 Objectives

The logbook assignment is given to year 2 students for Network Operating System module. This is the week 8 logbook assignment where students are required to create a multi-level directory and perform different operations using various command. It was done at the workshop of the module.

The objectives of this logbook are listed below:

- To learn how to create multi-level complex directory using "mkdir" with and without "p" option.
- To learn how to copy, move, and delete file between sub directories of complex directory.
- To practice changing directory using absolute and relative path.
- To learn different mode of listing files and their significance.
- To change access of read, write, and execute between different types of users.

1.3 Required tools and concepts

The tools and concepts required for are:

Tools

- Debian Linux operating system and its terminal to run the command.
- Mkdir to make complex nested directory.
- To change directory "cd" was used.
- To list directory/file "ls" was used.
- "cp" was used to copy files and "mv" to move files
- To delete files and directories "rm" and "rmdir" was used.
- "chmod " was used to remove files and directories permissions as well as to give permissions.
- "echo" was used to display the text.
- "cat" and "touch" was used to create file.

Concepts

- Pathname concept to change the directory.
- Hierarchy concept of directory to navigate for different file manipulation operations.
- File management concept to create, copy, move and delete files.
- Sub directories concept to manage directory in an efficient way.
- Access to files, read and write, was restricted through the concepts of file permission.

1.4 Steps required for lab

First, open Debian terminal in your host OS to run command of Linux and follow these steps:

Create directory structure

We can create directory using mkdir with or without -p option. Let's create it without using -p option.

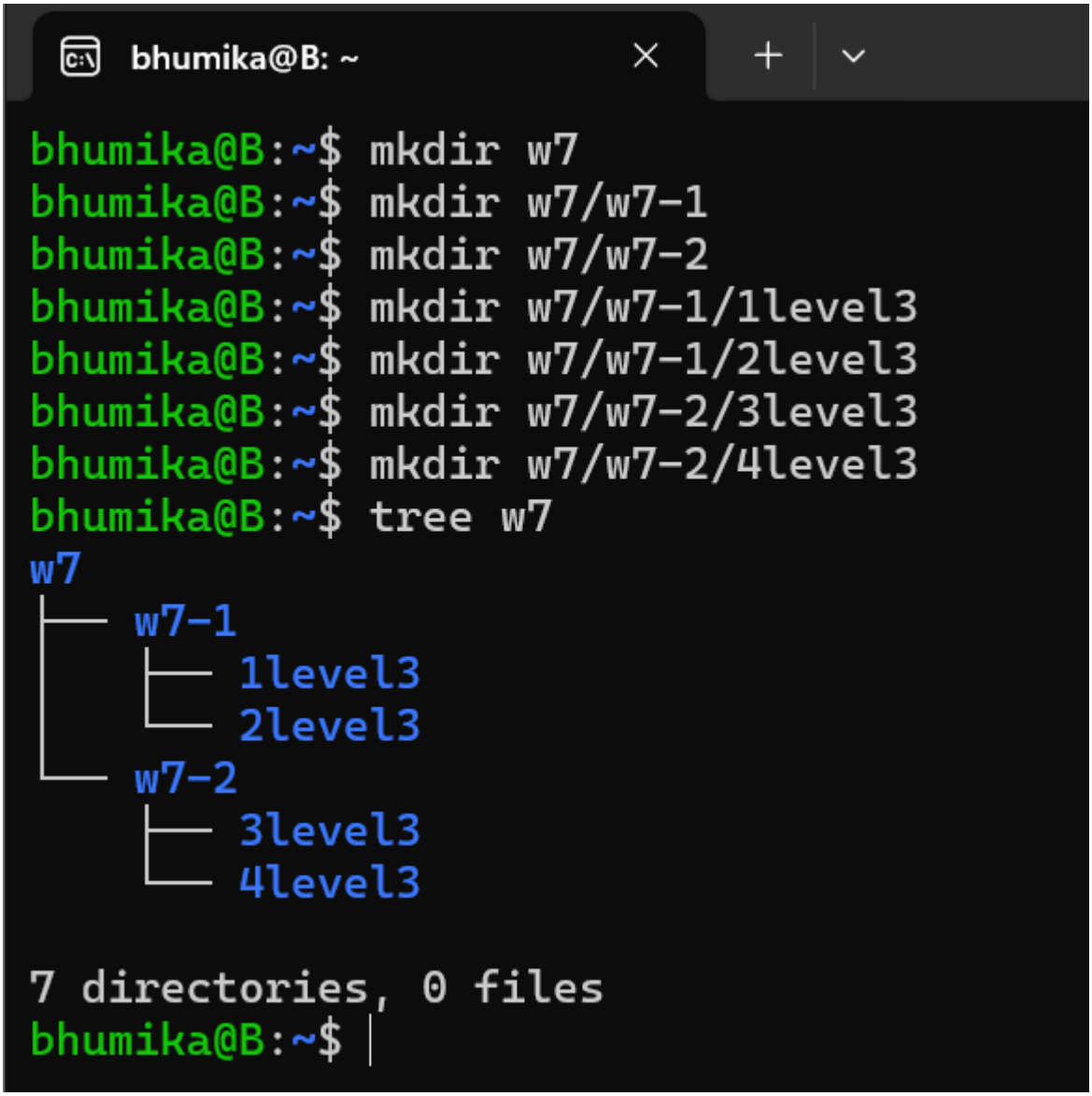

Figure 1: Creating directory structure

Now, creating directory with -p option. It enables you to create parent and its child directories in a single line.

```
bhumika@B:~$ mkdir -p w7/{w7-1/{1level3,2level3},w7-2/{3level3,4level3}}
bhumika@B:~$ tree w7
w7
├── w7-1
│   ├── 1level3
│   └── 2level3
└── w7-2
    ├── 3level3
    └── 4level3

7 directories, 0 files
bhumika@B:~$
```

Figure 2: Creating directory structure with -p option

Changing directory to 1level3

To change the directory using relative pathname, we have to write cd command and give pathname.

```
bhumika@B:~$ cd w7/w7-1/1level3
bhumika@B:~/w7/w7-1/1level3$
```

Figure 3: changing directory to 1level3

Practicing changing the directory using relative pathname

To change the directory from the 1level3 to 2level3 we have to use “..” to go back to the parent directory and from w7-1, we have to enter to the 2level3. Likewise, for going to 4level3, we again have to go back to w7 using “..” and then enter to the w7-2 to 4level3. Finally, for going to the w7 directory we have to again use “..” to return from 4level3.

```
bhumika@B:~/w7/w7-1/1level3$ cd ../2level3
bhumika@B:~/w7/w7-1/2level3$ cd ../../w7-2/4level3
bhumika@B:~/w7/w7-2/4level3$ cd ../..
bhumika@B:~/w7$
```

Figure 4: Changing directory

Creating file inside 1level3

First, we have to change the directory to 1level3 and create a text file inside it with the choice of your tool. I have created a text file named “bhumika” using cat.

```
bhumika@B:~$ cd w7/w7-1/1level3
bhumika@B:~/w7/w7-1/1level3$ cat > bhumika
This is file made inside
level 1 by bhumikabhumika@B:~/w7/w7-1/1level3$
bhumika@B:~/w7/w7-1/1level3$ tree /home/bhumika/w7
/home/bhumika/w7
├── w7-1
│   ├── 1level3
│   │   └── bhumika
│   └── 2level3
└── w7-2
    ├── 3level3
    └── 4level3

7 directories, 1 file
bhumika@B:~/w7/w7-1/1level3$
```

Figure 5: Creating a text file

Copying text file to different directories

To copy the text file, we have to use cp command and give the file name which we want to copy and the path.

```
bhumika@B:~/w7/w7-1/1level3$ cp bhumika file1
bhumika@B:~/w7/w7-1/1level3$ cp bhumika /home/bhumika/w7/w7-1/2level3/file1
bhumika@B:~/w7/w7-1/1level3$ cp bhumika ../../w7-2/3level3/file1
bhumika@B:~/w7/w7-1/1level3$ tree ~/w7
/home/bhumika/w7
├── w7-1
│   ├── 1level3
│   │   ├── bhumika
│   │   └── file1
│   └── 2level3
│       └── file1
└── w7-2
    ├── 3level3
    │   └── file1
    └── 4level3

7 directories, 4 files
bhumika@B:~/w7/w7-1/1level3$
```

Figure 6: Copying file to different directories

Moving text file from 1level3 to 4level3

To move the text file “file1” from 1level3 to 4level3, we have use command mv and give the file name which we want to move and the path.

```
bhumika@B:~/w7/w7-1/1level3$ mv file1 /home/bhumika/w7/w7-2/4level3/file1
bhumika@B:~/w7/w7-1/1level3$ tree ~/w7
/home/bhumika/w7
├── w7-1
│   ├── 1level3
│   │   └── bhumika
│   └── 2level3
│       └── file1
└── w7-2
    ├── 3level3
    │   └── file1
    └── 4level3
        └── file1

7 directories, 4 files
```

Figure 7: Moving file to another directory

Printing the text using echo

To print the text using echo, we have to write command “echo -e "Text to print"”.

```
bhumika@B:~/w7/w7-1/1level3$ echo -e "Hello! I can do it\n 5>(20:8)<(30*2) \n Line 1 \n Line 2 \n a-b,A-B,-,+,<,>,#,$,%,&. \n"
Hello! I can do it
 5>(20:8)<(30*2)
 Line 1
 Line 2
 a-b,A-B,-,+,<,>,#,$,%,&.
```

Figure 8: Printing the text using echo

Using ls command without and with different options

For W7 directory

When we use ls without any option, it lists basic visible file/folder which are on the current directory.

```
bhumika@B:~/w7/w7-1/1level3$ ls ../../
w7-1  w7-2
```

Figure 9: using ls in w7 directory

When we use ls with "a" option, it lists visible as well as hidden file.

```
bhumika@B:~/w7/w7-1/1level3$ ls -a ../../
.  ..  w7-1  w7-2
```

Figure 10: Using ls in w7 with option "a"

When we use ls with "d" option, it shows directory only not what is inside it.

```
bhumika@B:~/w7/w7-1/1level3$ ls -d ../../
../../
```

Figure 11: Using ls in w7 with option "d"

When we use ls with "g" option, it shows group information.

```
bhumika@B:~/w7/w7-1/1level3$ ls -g ../../
total 8
drwxr-xr-x 4 bhumika 4096 Dec 14 02:51 w7-1
drwxr-xr-x 4 bhumika 4096 Dec 14 02:51 w7-2
```

Figure 12: Using ls in w7 with option "g"

When we use ls with "l" option, it shows detailed information of the directory.

```
bhumika@B:~/w7/w7-1/1level3$ ls -l ../../
total 8
drwxr-xr-x 4 bhumika bhumika 4096 Dec 14 02:51 w7-1
drwxr-xr-x 4 bhumika bhumika 4096 Dec 14 02:51 w7-2
```

Figure 13: Using ls in w7 with option "l"

When we use ls with "R" option, it lists sub directories content as well.

```
bhumika@B:~/w7/w7-1/1level3$ ls -R ../../
../../:
w7-1  w7-2

../../w7-1:
1level3  2level3

../../w7-1/1level3:
bhumika

../../w7-1/2level3:
file1

../../w7-2:
3level3  4level3

../../w7-2/3level3:
file1

../../w7-2/4level3:
file1
bhumika@B:~/w7/w7-1/1level3$
```

Figure 14: Using ls in w7 with option "R"

For W7-1 directory

When we use ls without any option, it lists basic visible file/folder which are on the current directory.

```
bhumika@B:~/w7/w7-1/1level3$ ls ../
1level3  2level3
```

Figure 15: Using ls in w7-1 directory

When we use ls with "a" option, it lists visible as well as hidden file.

```
bhumika@B:~/w7/w7-1/1level3$ ls -a ../
.  ..  1level3  2level3
```

Figure 16: Using ls in w7-1 with option "a"

When we use ls with "d" option, it shows directory only not what is inside it.

```
bhumika@B:~/w7/w7-1/1level3$ ls -d ../
../
```

Figure 17: Using ls in w7-1 with option "d"

When we use ls with "g" option, it shows group information.

```
bhumika@B:~/w7/w7-1/1level3$ ls -g ../
total 8
drwxr-xr-x 2 bhumika 4096 Dec 14 03:16 1level3
drwxr-xr-x 2 bhumika 4096 Dec 14 03:09 2level3
```

Figure 18: Using ls in w7-1 with option "g"

When we use ls with "l" option, it shows detailed information of the directory.

```
bhumika@B:~/w7/w7-1/1level3$ ls -l ../
total 8
drwxr-xr-x 2 bhumika bhumika 4096 Dec 14 03:16 1level3
drwxr-xr-x 2 bhumika bhumika 4096 Dec 14 03:09 2level3
```

Figure 19: Using ls in w7-1 with option "l"

When we use ls with "R" option, it lists sub directories content as well.

```
bhumika@B:~/w7/w7-1/1level3$ ls -R ../
../:
1level3  2level3

../1level3:
bhumika

../2level3:
file1
bhumika@B:~/w7/w7-1/1level3$
```

Figure 20: Using ls in w7-1 with option "R"

For 1level3 directory

When we use ls without any option, it lists basic visible file/folder which are on the current directory.

```
bhumika@B:~/w7/w7-1/1level3$ ls
bhumika
```

Figure 21: Using ls in 1level3 directory

When we use ls with "a" option, it lists visible as well as hidden file.

```
bhumika@B:~/w7/w7-1/1level3$ ls -a
.  ..  bhumika
```

Figure 22: Using ls in 1level3 with option "a"

When we use ls with "d" option, it shows directory only not what is inside it.

```
bhumika@B:~/w7/w7-1/1level3$ ls -d
.
```

Figure 23: Using ls in 1level3 with option "d"

When we use ls with “g” option, it shows group information.

```
bhumika@B:~/w7/w7-1/1level3$ ls -g
total 4
-rw-r--r-- 1 bhumika 43 Dec 14 03:04 bhumika
```

Figure 24: Using ls in 1level3 with option “g”

When we use ls with “l” option, it shows detailed information of the directory.

```
bhumika@B:~/w7/w7-1/1level3$ ls -l
total 4
-rw-r--r-- 1 bhumika bhumika 43 Dec 14 03:04 bhumika
```

Figure 25: Using ls in 1level3 with option “l”

When we use ls with "R" option, it lists sub directories content as well.

```
bhumika@B:~/w7/w7-1/1level3$ ls -R
.:
bhumika
```

Figure 26: Using ls in 1level3 with option "R"

Removing w7-2 directory and its sub directories

First, we have to change the directory to "w7".

```
bhumika@B:~/w7/w7-1/1level3$ cd ../../
bhumika@B:~/w7$ rm -r -i w7-2
```

Figure 27: Changing directory to w7

Now, to remove the w7-2 directory and its sub directories we have to use command "rm -r -i w7-2". The option "r" removes content recursively and "i" ask for permissions before deleting each of the directory/file/folder.

```
bhumika@B:~/w7$ rm -r -i w7-2
rm: descend into directory 'w7-2'? y
rm: descend into directory 'w7-2/3level3'? y
rm: remove regular file 'w7-2/3level3/file1'? y
rm: remove directory 'w7-2/3level3'? y
rm: remove directory 'w7-2'? y
bhumika@B:~/w7$ tree
.
└── w7-1
    ├── 1level3
    │   └── bhumika
    └── 2level3
        └── file1

4 directories, 2 files
bhumika@B:~/w7$
```

Figure 28: Removing w7-2

Changing to w7-1 directory and performing these operations

First changing the directory to w7-1.

```
bhumika@B:~/w7$ cd w7-1
bhumika@B:~/w7/w7-1$ ls -l 1level
```

Figure 29: Changing to w7-1 directory

Displaying access permission of bhumika inside w7-1

To display the permissions of bhumika, we have to write command "ls -l 1level1".

```
bhumika@B:~/w7/w7-1$ ls -l 1level3
total 4
-rw-r--r-- 1 bhumika bhumika 43 Dec 14 03:04 bhumika
bhumika@B:~/w7/w7-1$
```

Figure 30: Displaying the access permissions

Removing and displaying access permissions for bhumika

To remove the access permissions for bhumika, we have to write the command "chmod -rw 1level3/bhumika" and to display the permissions we have to write the command "ls -l 1level3".

```
bhumika@B:~/w7/w7-1$ chmod -rw 1level3/bhumika
bhumika@B:~/w7/w7-1$ ls -l 1level3
total 4
---------- 1 bhumika bhumika 43 Dec 14 03:04 bhumika
bhumika@B:~/w7/w7-1$
```

Figure 31: Removing the access permission

Reading the file using cat utility

To read the file, we have to use command "cat 1level3/bhumika". However, we cannot read the file because the permission to read is removed.

```
bhumika@B:~/w7/w7-1$ cat 1level3/bhumika
cat: 1level3/bhumika: Permission denied
bhumika@B:~/w7/w7-1$
```

Figure 32: Reading the file

Writing into the file using cat utility

To write in the file, we have to use command "cat >> 1level3/bhumika". However, we cannot write in the file because the permission to write is removed.

```
bhumika@B:~/w7/w7-1$ cat >> 1level3/bhumika
-bash: 1level3/bhumika: Permission denied
bhumika@B:~/w7/w7-1$
```

Figure 33: Writing in the file

Adding read and write permissions to the user and displaying the permissions

To add read and write permissions, we have to write command " chmod 600 1level3/bhumika" and we have to write command " ls -l 1level3/bhumika" to display the permissions.

```
bhumika@B:~/w7/w7-1$ chmod 600 1level3/bhumika
bhumika@B:~/w7/w7-1$ ls -l 1level3/bhumika
-rw------- 1 bhumika bhumika 43 Dec 16 02:30 1level3/bhumika
bhumika@B:~/w7/w7-1$
```

Figure 34: Adding and displaying the permissions

Reading the file using cat utility after giving permission

To read the file, we have to use command "cat 1level3/bhumika".

```
bhumika@B:~/w7/w7-1$ cat 1level3/bhumika
This is file made inside
level 1 by bhumikabhumika@B:~/w7/w7-1$
```

Figure 35: Reading the file after giving permission

Writing into the file using cat utility after giving permission

To write in the file, we have to use command "cat >> 1level3/bhumika".

```
bhumika@B:~/w7/w7-1$ cat >> 1level3/bhumika
I am back hahabhumika@B:~/w7/w7-1$
bhumika@B:~/w7/w7-1$
```

Figure 36: Writing in the file after giving permission

Displaying what is written inside the file.

```
bhumika@B:~/w7/w7-1$ cat 1level3/bhumika
This is file made inside
level 1 by bhumikaI am back hahabhumika@B:~/w7/w7-1$
```

Figure 37: Displaying the file where we made change

Displaying the file permissions of 1level3

To display the file permissions, we have to write the command " ls -l".

```
bhumika@B:~/w7/w7-1$ ls -l
total 8
drwxr-xr-x 2 bhumika bhumika 4096 Dec 14 03:16 1level3
drwxr-xr-x 2 bhumika bhumika 4096 Dec 14 03:09 2level3
bhumika@B:~/w7/w7-1$
```

Figure 38: Displaying the file permissions

Removing and displaying all access permissions for 1level3

To remove the permissions, we have to write command "chmod -rwx 1level3"and to display the permissions we have to write command " ls -l".

```
bhumika@B:~/w7/w7-1$ chmod -rwx 1level3
bhumika@B:~/w7/w7-1$ ls -l
total 8
d--------- 2 bhumika bhumika 4096 Dec 14 03:16 1level3
drwxr-xr-x 2 bhumika bhumika 4096 Dec 14 03:09 2level3
bhumika@B:~/w7/w7-1$
```

Figure 39: Removing and displaying access permission

Reading the file from 1level3

To read the file from 1level3, we have to write the command "cat 1level3/file1". However, we cannot read the file because permission to read is removed.

```
bhumika@B:~/w7/w7-1$ cat 1level3/file1
cat: 1level3/file1: Permission denied
```

Figure 40: Reading the file

Putting file in the 1level3

To put a file inside 1level3 directory, we have to use command "touch 1level3/file2". However, we cannot do so because the permission to do so is removed.

```
bhumika@B:~/w7/w7-1$ touch 1level3/file2
touch: cannot touch '1level3/file2': Permission denied
bhumika@B:~/w7/w7-1$
```

Figure 41: Creating a file using touch

Listing file inside 1level3

To list the files from 1level3, we have to use the command "ls -l". However, we cannot do so because we don't have permission to list what is inside 1level3 directory.

```
bhumika@B:~/w7/w7-1$ ls 1level3
ls: cannot open directory '1level3': Permission denied
bhumika@B:~/w7/w7-1$
```

Figure 42: Listing what is inside 1level3

Adding read, write, and execute access to the user

To add the read, write, and execute access to the user, we have to write command "chmod 700 1level3".

```
bhumika@B:~/w7/w7-1$ chmod 700 1level3
bhumika@B:~/w7/w7-1$ ls -l
total 8
drwx------ 2 bhumika bhumika 4096 Dec 14 03:16 1level3
drwxr-xr-x 2 bhumika bhumika 4096 Dec 14 03:09 2level3
bhumika@B:~/w7/w7-1$
```

Figure 43: Adding read, write, and execute access

Displaying access permissions of 1level3

To display access permissions of 1level3, we have to write the command "ls -l".

```
bhumika@B:~/w7/w7-1$ ls -l
total 8
drwx------ 2 bhumika bhumika 4096 Dec 14 03:16 1level3
drwxr-xr-x 2 bhumika bhumika 4096 Dec 14 03:09 2level3
bhumika@B:~/w7/w7-1$
```

Figure 44: Displaying the access permission of 1level3

Reading file from 1level3

To read a file from 1level3, we have to write the command "cat 1level3/bhumika".

```
bhumika@B:~/w7/w7-1$ cat 1level3/bhumika
This is file made inside
level 1 by bhumikabhumika@B:~/w7/w7-1$
```

Figure 45: Reading file from 1level3 after giving access to user

Creating file inside 1level3

To create a file inside 1level3, we have to write the command "touch 1level3/file2".

```
bhumika@B:~/w7/w7-1$ touch 1level3/file2
bhumika@B:~/w7/w7-1$ ls 1level3
bhumika  file2
bhumika@B:~/w7/w7-1$
```

Figure 46: Creating file inside 1level3

Searching file in 1level3 using ls command

To search a file inside 1level3, we have to use the command "ls 1level3".

```
bhumika@B:~/w7/w7-1$ ls 1level3
bhumika  file2
bhumika@B:~/w7/w7-1$
```

Figure 47: Searching file inside 1level3

1.5 Conclusion

The lab focused on creating complicated directory hierarchies with the help of "mkdir" with the standard and -p flags and moving around several levels of directories using relative paths and parent directories (..). It covered holistic file tasks such as transferring files to other destinations, transferring the files between directories, and deleting files safely with interactive messages.

The practical knowledge provided me with important information on the variations of ls commands that display hidden files (a), directories alone (d), group information (g), detailed listings (l), and recursive (R). The "chmod" permission testing indicated how the removal/restoration of "rwx" allowed access to be blocked until it was fully restored, which impeded reading, writing and navigation of a directory.

The workshop demonstrated real implications of security and the behaviour of the commands in the true UNIX setups through its extensive tests on file and folder access following permission changes. "Echo" and "printf" showed basic output display mechanism.

Overall, this lab helped me in gaining holistic UNIX administration skills to include directory management, file management, permission security and command customization that is critical to professional Linux system and network activities.